# Conditions and grouped assumptions

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LLM-assisted syntax specimen; see PROVENANCE.md.

## Repeated assumptions

**Result 1 (symmetry restated).** [Clean condition examples, symmetry restated]
Let $n$ be a natural number. Let $A$ and $B$ be $n \times n$ matrices over $\mathbb{R}$. Assume $A$ is symmetric (`symmetry of A`). Assume $B$ is symmetric (`symmetry of B`). Then

$$A \text{ is symmetric and } B \text{ is symmetric.}$$

**Result 2 (positive values).** [Clean condition examples, positive values]
Let $x$ and $y$ be real numbers. Assume $x > 0$ (`positivity of x`). Assume $y > 0$ (`positivity of y`). Then

$$x > 0 \text{ and } y > 0.$$

## Explicit context references

**Context 1 (symmetric matrices).**
Let $n$ be a natural number. Let $A$ and $B$ be $n \times n$ matrices over $\mathbb{R}$. Assume $A$ is symmetric (`symmetry of A`). Assume $B$ is symmetric (`symmetry of B`).

**Result 3 (symmetry restated).** [Clean condition examples, symmetry restated]
Using Context (1). Then

$$A \text{ is symmetric and } B \text{ is symmetric.}$$

**Result 4 (positive values).** [Clean condition examples, positive values]
Let $x$ and $y$ be real numbers. Assume $x > 0$ (`positivity of x`). Assume $y > 0$ (`positivity of y`). Then

$$x > 0 \text{ and } y > 0.$$